A 98.6%-Efficient 3.3-kW SiC Totem-Pole PFC Rectifier With Adaptive Dead-Time Control

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Abstract—This paper presents a 3.3-kW bridgeless totem-pole rectifier built with 1200-V silicon carbide MOSFETs. An adaptive dead-time controller cuts the light-load losses. The prototype reaches a peak efficiency of 98.6% at 50% load and a power factor above 0.99 from 20% load upward, at 230-V input.

I. INTRODUCTION

Power factor correction (PFC) is required for power supplies above 75 W connected to the public grid. The boost rectifier with a diode bridge remains common, but the bridge conducts the full input current through two diodes, and its loss limits the efficiency to about 97% at this power level.

Bridgeless topologies remove the diode bridge. Among them, the totem-pole rectifier needs the fewest components, but its fast leg operates in continuous conduction mode only with switches whose body diode has a negligible reverse recovery charge. Silicon carbide (SiC) MOSFETs meet this condition, and soft switching through zero-voltage switching (ZVS) reduces their turn-on loss further.

Prior work reported peak efficiencies between 98.0% and 98.4% at comparable power levels [3], [7]. In those designs the dead time between the two switches of the fast leg is fixed, and it is set for the worst case. At light load, the switch node then rings for most of the dead time, and the body diode conducts. This paper adapts the dead time to the load current and measures the effect on efficiency and on total harmonic distortion (THD).

II. CONVERTER DESIGN

The fast leg uses two KV1200M040 SiC MOSFETs, whose on-state resistance is $R_{DS(on)} = 40\ \text{m}\Omega$ at $T_j = 25\ ^\circ\text{C}$, rising to 68 mΩ at $T_j = 175\ ^\circ\text{C}$. The junction-to-case thermal resistance is $R_{\theta JC} = 0.38$ K/W. The slow leg uses two 600-V silicon superjunction MOSFETs switching at the line frequency.

The boost inductor was designed for a current ripple of 20% of the peak line current at full load, which gives $L = 450\ \mu\text{H}$. The switching frequency is $f_{sw} = 65$ kHz, chosen to keep the second harmonic of the switching frequency below the 150-kHz start of the conducted emission band.

The conduction loss of each fast-leg switch is $P_{cond} = I_{rms}^2 R_{DS(on)}$, evaluated at the junction temperature reached at each load. The gate is driven between -4 V and +18 V. The drain leakage current at 800 V stayed below 2.5×10^{-6} A over the whole temperature range.

III. ADAPTIVE DEAD-TIME CONTROL

The controller measures the inductor current at each switching period and selects the dead time from a table indexed by current. At high current the switch node transition is fast and a short dead time suffices; at low current the transition is slow, and too short a dead time loses ZVS. The table spans 40 ns to 120 ns, against a fixed dead time of 150 ns in the reference design.

The controller runs in the same microcontroller as the current loop and needs no extra sensing: the current is already sampled for the current loop. Its only cost is a table of 32 entries.

IV. EXPERIMENTAL RESULTS

The prototype was tested at 230-V, 50-Hz input and 400-V output. The converter was tested over an ambient temperature range of $-25\ ^\circ\text{C}$ to $55\ ^\circ\text{C}$ in a climatic chamber. Table I gives the measured performance at 25 °C ambient.

TABLE I
MEASURED PERFORMANCE AT 230-V INPUT

Load	Efficiency	Power factor	THD
20%	97.1%	0.991	6.8%
50%	98.6%	0.998	3.1%
100%	98.2%	0.999	2.4%

The adaptive dead time reduced the light-load losses by 18% compared with the fixed dead time of 150 ns. At 20% load, the efficiency improved from 96.4% to 97.1%. At full load the dead time made no measurable difference, since the transition is fast in both cases.

ZVS was achieved above 35% load with the adaptive dead time, and above 60% load with the fixed one. Below these loads the switches turn on with a partly discharged output capacitance.

The THD stayed below 10% from 10% load upward, and below 5% from 40% load upward, which meets the limits of IEC 61000-3-2 for class A equipment. The measured power factor exceeded 0.99 from 20% load upward.

At full load and 55 °C ambient, the heatsink temperature reached 71 °C and the junction temperature of the fast-leg switches was estimated at 112 °C from the measured case temperature and $R_{\theta JC}$.

V. DISCUSSION

The gain at light load matters most for server power supplies, which spend most of their life between 20% and 50% load. A further gain is possible by lowering f_{sw} at light load, which this work did not attempt, because it moves the inductor ripple outside its design range.

The thermal margin at 55 °C allows operation at full power without forced derating, although the fan speed must then be at its maximum. The acoustic noise at that speed was not measured.

VI. CONCLUSION

The proposed converter exceeds the 80 PLUS Titanium requirement of 96% efficiency at 50% load with a margin of 2.6 percentage points. The adaptive dead time costs no hardware and should apply to any totem-pole rectifier with wide-bandgap switches.

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